

Proprietary Material

Information and descriptions contained in this manual are the property of Bistos Corporation and may not be copied, reproduced, disseminated, or distributed without express written permission from Bistos Corporation.

Information furnished by Bistos Corporation is believed to be accurate and reliable. However, no responsibility is assumed by Bistos for its use, or any infringements of patents or other rights of third parties that may result from its use. No license is granted by implication or otherwise under any patent or patent rights of Bistos

Revision 05
October, 2014

Copyright © Bistos Corporation 2014. All rights reserved.

7th Fl., A Bldg., Woolim Lions Valley 5-cha,
302, Galmachi-ro, Seongnam-si, Gyeonggi-do,
Korea

Telephone: ++82 31 750 0340
Fax: ++82 31 750 0344

Printed in Korea

Table of Contents

Proprietary Material	2
Section 1 Safety	4
1.1. Instructions for the Safe Operation and Use of the BT-350 Monitor	4
1.2. Warnings	5
1.3. Cautions	5
1.4. General Precaution on Environment	7
1.5. General Precaution on Electric Safety	8
1.6. Definition of Symbols	9
Section 2 Description of Product	10
2.1. Product Overview	10
2.2. Intended USE	10
2.3. Product Features	10
2.4. Exterior Component Designations	11
2.5. Specifications	14
Section 3 Disassembly & Assembly	16
3.1. Product	16
3.2. LCD Ass'y	19
3.3. Doppler Probe	20
3.4. UC Probe	20
3.5. Connector Position	21
Section 4 Part List	22
4.1. Top Ass'y Parts	22
4.2. Bottom Ass'y Parts	23
4.3. Accessory & Option Parts	24
Section 5 External device Communication	25
5.1. RS-232C	25
Section 6 Software Upgrade	26
6.1. How to do S3c2410 flash fusing	26

Section 7 Product Setting	32
7.1. Power & Accessories Connection	32
 Section 8 Maintenance & Conservation	 33
8.1. Monitor	33
8.2. Transducers	33
8.3. Belts	34
8.4. Contacting components and characteristics	34
8.5. Description of Cidex TM	34
8.6. Battery Disposal and Handling	34
8.7. Maintenance	35
8.8. Disposal of the BT-350	35
8.9. Request a service for general problems	35
 Section 9 Troubleshooting	 36
9.1. Warning Sign Occasion and Status Checking	36
9.2. Troubleshooting on the Other Situation	38
9.3. Technical Descriptions	40

Section 1

Safety

1.1 . *Instructions for the Safe Operation and Use of the BT-350 Monitor*

- Examine the monitor and any accessories periodically to ensure that the cables, line cords, transducers, and instruments do not have visible evidence of damage that may affect patient safety or monitoring performance. The recommended inspection interval is once per week or less. Do not use the monitor if there is any visible sign of damage.
- Only the AC line cord supplied with the BT-350, or its equivalent, is approved for use with the Unit.
- Do not attempt to service the BT-350 monitor. Only qualified service personnel by Bistos Co.,Ltd. should attempt any needed internal servicing.
- The BT-350 is not specified or intended for operation during the use of defibrillators or during defibrillator discharge.
- The BT-350 is not specified or intended for operation in the presence of electrosurgical equipment.
- The BT-350 is not specified or intended for operation in conjunction with any other type of monitoring equipment except the specific devices that have been identified for use in this Operator's Manual.
- Perform periodic safety testing to insure proper patient safety. This should include leakage current measurement and insulation testing. The recommended testing interval is once per year.
- Do not operate the BT-350 monitor if it fails to pass the power on self-test procedure.

WARNING: Be informed that it may cause serious injury or death to the patient, property damage, material losses against the "Warning" sign.

CAUTION: Be informed that it may cause no harm in life but lead to injury against the "Caution" sign.

1.2. Warnings

WARNING: EXPLOSION HAZARD — Do not use the BT-350 in a flammable atmosphere where concentrations of flammable anesthetics or other materials may occur.

WARNING: SHOCK HAZARD — The power receptacle must be a three wire grounded outlet. Never adapt the three-prong plug to fit a two-slot outlet. If the outlet has only two slots, make sure that it is replaced with a three-slot grounded outlet before attempting to operate the monitor.

WARNING: Do not connect to an electrical outlet controlled by a wall switch.

WARNING: SHOCK HAZARD — Do not attempt to connect or disconnect a power cord with wet hands. Make certain that your hands are clean and dry before touching a power cord.

WARNING: Use only patient cables and transducers supplied with the monitor. Use of any other patient cables may result in out-of-specification performance and possible safety hazards.

WARNING: Do not contact RS-232C port and patient at the same time.

WARNING: AC/DC Adaptor should use appointed product.

WARNING: SHOCK HAZARD — Do not attempt to disjoint the power adaptor exterior with no permission. It may cause electric shock. Also it has low possibility of reaching to death. In the case of you have some problems with the power adaptor, we recommend that you have to contact to us first of all.

WARNING: SHOCK HAZARD — Do not touch the patient simultaneously with contacting signal connector, other equipment or ground. This can cause the electric shock to the patient or operator.

WARNING: SHOCK HAZARD — During upgrading the BT-350, do not use the BT-350 to the patient. This can cause the electric shock to the patient.

1.3. Cautions

CAUTION:

- The equipment conforms to Class A according to IEC/EN 60601-1(Safety of Electric Medical Equipment)
- This equipment conforms to Level B according to IEC/EN 60601-1-2 (Electromagnetic Compatibility Requirements)

CAUTION: The relevant law restricts this device to sale by or on the order of a physician.

CAUTION: Keep the operating environment free of dust, vibrations, corrosive, or flammable materials, and extremes of temperature and humidity. The unit should be kept clean and free of transducer gel and other substances.

CAUTION: When installing the unit into a sealed place, allow for adequate ventilation, accessibility for servicing, and room for adequate visualization and operation. Two side of the device is must be opened.

CAUTION: Do not operate the unit if it is damp or wet because of condensation or spills. Avoid using the equipment immediately after moving it from a cold environment to a warm, humid location.

CAUTION: Never use sharp or pointed objects to operate the front-panel switches.

CAUTION: General-purpose personal computers and modems are not designed to meet the electrical safety requirements of medical devices. The RS-232C connector on the BT-350 is electrically isolated to permit safe connections to non-medical devices, which should be connected with a cable of sufficient length to prevent the non-medical equipment from contacting the patient. If the BT-350 have to be connected another medical devices, it must be complied with the standards IEC/EN 60601-1 and IEC/EN 60601-1-2.

CAUTION: Do not autoclave or gas sterilize the monitor or any accessories. Follow cleaning and disinfection instructions in Section 9 of this manual.

CAUTION: Do not immerse BT-350 main body and trasducers in liquid. When using solutions, use sterile wipes to avoid pouring fluids directly on the transducer. Follow cleaning and disinfection instructions in Section 9 of this manual.

CAUTION: When washing the transducer belts, the water temperature must not exceed 60°C (140°F).

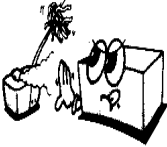
CAUTION: When loading paper, the paper must be put above the shaft. Otherwise, the paper can be biased one side.

CAUTION: If the equipment use in area where the integrity of the external protective conductor in the installation or its arrangement is in doubt, equipment shall be operated from its internal electrical source when the optional battery is selected.

CAUTION: When the printer door is open, do not put the finger to the inside of BT-350. This can cause the finger wound. Also do not prick the inside of BT-350 when the printer door is open. This can cause the damage to the device or electric shock.

1.4 General Precaution on Environment

- Do not keep or operate the equipment under the environment listed below.

	Avoid placing in an area exposed to moisture. Do not touch the equipment with wet hand.		Avoid exposure to direct sunlight
	Avoid placing in an area where there is a high variation of temperature. Operating temperature ranges from 10°C to 40°C. Operating humidity ranges from 20% to 90%.		Avoid in the vicinity of Electric heater
	Avoid placing in an area where there is an excessive humidity rise or ventilation problem.		Avoid placing in an area where there is an excessive shock or vibration.
	Avoid placing in an area where chemicals are stored or where there is in danger of gas leakage.		Avoid dust and especially metal material into the equipment.
	Do not disjoint or disassemble the equipment. BISTOS Co., Ltd. does not take responsibility of it.		Power off when the equipment is not fully installed. Otherwise, the equipment could be damaged.

1.5. General Precaution on Electric Safety

- Check the items listed below before operating the equipment.
 - Be sure that power supply line is appropriate to use.
(Power Adaptor Input: 100 ~ 240V AC, Power Adaptor Output: 18V, 3.4A).
 - Be sure that the entire connection cable of the system is properly and firmly fixed.

Note

BT-350 includes BT-350LCD and BT-350LED.

Note

The equipment should not be placed in the vicinity of electric generator, X-ray, broadcasting apparatus to eliminate the electric noise during operation. Otherwise, it may cause incorrect result.
Self-power line is important for BT-350. To use the same power source with other electric instruments may cause incorrect result.

Note

BT-350 is classified as listed below;

- This equipment conforms to Class I, Type-BF.
- Do not use the equipment in the vicinity of flammable anesthetics and solvents.
- The equipment conforms to Class I according to IEC/EN 60601-1 (Safety of Electric Medical Equipment)
- This equipment conforms to Level B according to IEC/EN 60601-1-2 (Electromagnetic Compatibility Requirements)

Note

Accessory equipment connected to the analog and digital interfaces must be certified according to the respective IEC standards (e.g. IEC 950 for data processing equipment and IEC 60601-1 for medical equipment). Furthermore all configurations shall comply with the system standard IEC 60601-1:2005.

If in doubt, consult the technical service department or your local representative.

1.6. Definition of Symbols

- The following is a list of symbols used on products manufactured by BISTOS CO.,LTD. Some symbols may not appear on your unit.

Symbol	Description	Requirements
	Power On/Off Button	IEC TR 60878
	This symbol identifies a safety note. Ensure you understand the function of this control before using it. There are no noted or identified hazards by ultrasound. But there is unknown hazardous possibility by ultrasound.	ISO 7000-0434A
	External Signal IN/OUT Port	IEC TR 60878
	Type BF Equipment	IEC 60417-5333
IPX8	IPX8 Waterproof (1 meter of water for 40 minutes.)	IEC60529
	Operating instructions	IEC7000-1641
	When disposing of some components (ex: internal NiMH battery), do not dispose as general wastes. Adhere to all applicable laws regarding recycling.	WEEE

※ According to IEC 60601-1-6 General requirements for basic safety and essential performance – Collateral Standard : Usability, the definition and using these symbols is adjusted.

Section 2

Description of Product

2.1. *Product Overview*

BT-350 is the fetal monitor that measures the fetal heart rate(FHR) which may be evaluated to predict fetal status and uterine contraction. BT-350 irradiates ultrasound wave to the abdomen of a pregnant woman, and detects the Doppler frequency signal reflected from the heart of the fetus. BT-350 analyzes this signal and displays the heart rate by color TFT-LCD(BT-350LCD) or 7segment led(BT-350LED). Also, BT-350 provides the sound from the heart of fetus. BT-350 measures the uterine contraction of a pregnant woman by pressure sensors and displays the numerical values. And BT-350 prints the heart rate of the fetus and the values of uterine contraction.

2.2. *Intended Use*

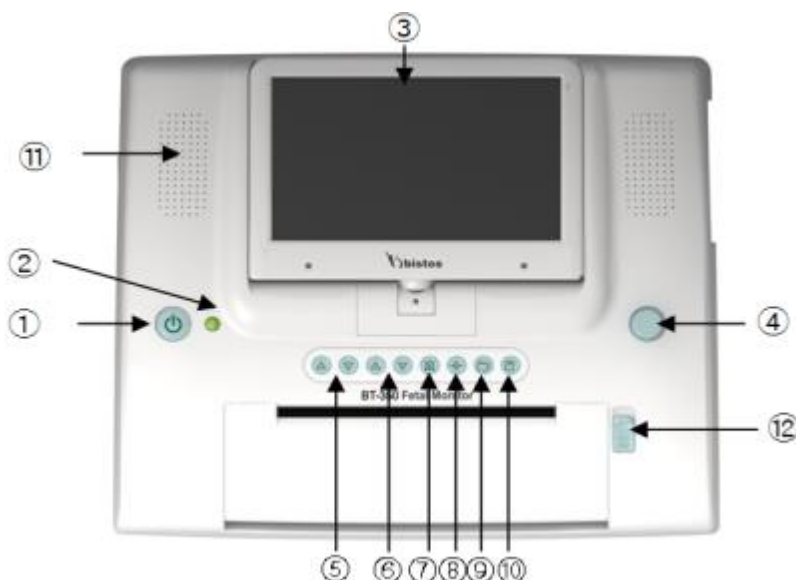
The BT-350 is a Prenatal Monitoring System for non-invasively measuring and showing graphically maternal abdominal contractions and the fetal heart rate by means of display on a non-permanent graphical display and on a strip chart recorder. This data is intended to aid in assessing the well being of the fetus during the final trimester of pregnancy (Non-Stress Test). This device is for use only by trained medical personnel located in hospitals, clinics, doctor's offices and in the patient's home.

2.3. *Product Features*

The monitored data can be recorded continuously or intermittently on a strip chart recorder at the operator's discretion. The recorded information includes graphic trend data and text information of monitor hardware and software configuration, date and time, patient identification, changes to operational settings, clinician and patient event marks.

2.4. Exterior Component Designations

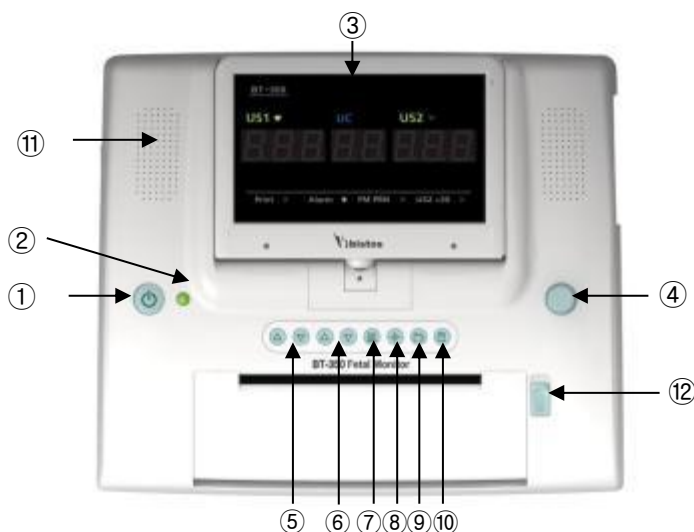
2.4.1. Description of the BT-350LCD Front Panel



[Figure 1.1 Front View of BT-350LCD]

- ① Power On/Off Button
- ② Power Indicating LED (AC:Green / Battery:Orange)
- ③ TFT-Color LCD
- ④ Control Knob
- ⑤ Dop1 Volume Up/Down Button
- ⑥ Dop2 Volume Up/Down Button
- ⑦ Alarm Sound On/Off Button
- ⑧ UC Zeroing Button
- ⑨ Mode Change Button
- ⑩ Printer On/Off Button
- ⑪ Speaker
- ⑫ Print Door Open Button

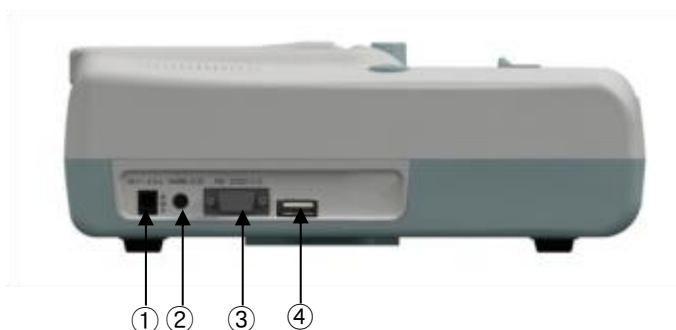
2.4.2. Description of the BT-350LED Front Panel



[Figure 1.2 Front View of BT-350LED]

- ① Power On/Off Button
- ② Power Indicating LED (AC:Green / Battery:Orange)
- ③ 7Segment LED Window
- ④ Control Knob
- ⑤ Dop1 Volume Up/Down Button
- ⑥ Dop2 Volume Up/Down Button
- ⑦ Alarm Sound On/Off Button
- ⑧ UC Zeroing Button
- ⑨ Setup Button
- ⑩ Printer On/Off Button
- ⑪ Speaker
- ⑫ Print Door Open Button

2.4.3. Description of the Left Panel



[Figure 1.3 Left View]

- ① Event Marker Connector
- ② RS-232C Port Connector
- ③ Power Adaptor Jack Connector
- ④ USB Port Connector

2.4.4. Description of the Right Panel



[Figure 1.4 Right View]

- ① DOP1/AST Connector
- ② DOP1/AST Connector
- ③ UC Connector

Warning

To avoid an expected electric shock, do not open the equipment cover or disassemble the equipment. Refer servicing to qualified personnel of BISTOS Co., Ltd.

2.5. Specifications

BT-350 Monitor Specifications:

2.5.1. Physical Characteristics

Dimensions – 9.6 cm H x 32.6 cm W x 27.6 cm D

Weight - approx. 5.5 kg

2.5.2. Safety

Complies with IEC 60601-1, IEC 60601-1-2

Class I Equipment & Internal Powered Equipment

Continuous Operation

Type BF applied parts

Dop / UC Probe : IPX8

2.5.3. Power

External:	Power Adaptor	Input : AC (100-240V), 50~60Hz, 1.5A Output: DC(18V), 3.4A
Internal:		NiMH, rechargeable battery 5 hours to full recharge during monitoring 5 hours to full recharge when monitor is off.
Power	AC -powered:	80VA, maximum
Dissipation:	Battery - powered	80VA, maximum Dop1/2 On[Vol4], Print On [3cm/min, FM Print On] : about 300minutes Dop1/2 On[Vol4], Print Off : about 300Hours

2.5.4. Environmental

Operating Temperature: 10°C to 40°C (50°F to 104°F)

Storage Temperature: –20°C to 60°C (–4°F to 140°F)

Relative Humidity: 20% to 90% non-condensing

Altitude: 0 -3048m (0 -10,000 ft)

2.5.5. Doppler Ultrasound FHR Monitoring

Parameter	Value
BPM Range:	30-240 BPM
Accuracy:	±2% of range
Leakage:	<10 μ A @ 264 VAC applied to transducer
Isolation:	>4 kV RMS, Type BF applied part

2.5.6. Uterine Contraction (TOCO) Monitoring

Parameter	Value
UC Range:	0-99 relative units
Resolution:	1 Count
Accuracy:	±1% relative unit
Leakage:	<10 μ A @ 264 VAC applied to transducer
Isolation:	>4 kV RMS, Type BF applied part

2.5.7. Paper

Pack Style:	Z-Fold.
Pack Size:	150 mm x 90 mm x 15 mm
End-of-Pack:	Mark along paper edge
Loading:	Open-door, slide-in
Paper Detectors:	Paper Out Loading Door Open

2.5.8. Paper Speeds

Normal: 1, 2, and 3 cm/min $\pm$ 1%
High-Speed: 10 cm/min (only in Trend mode)

Paper Tracking Accuracy: $\pm$ 1% (exclusive of paper accuracy)

Section 3

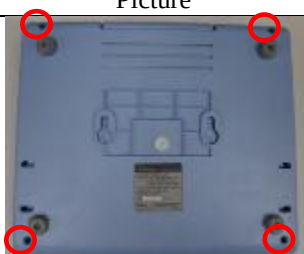
Disassembly & Assembly

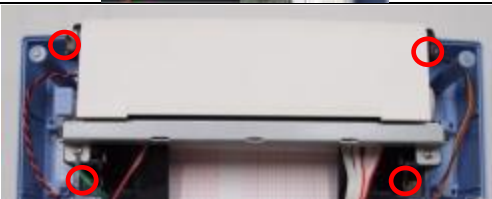
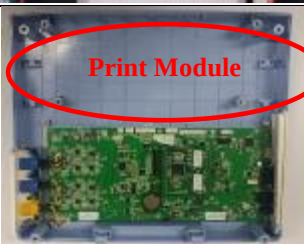
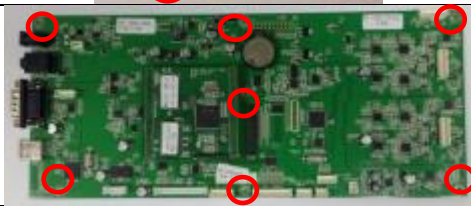
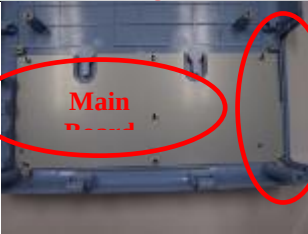
This Service Guide contains technical details for the BT-350 Fetal Monitor. It provides a technical foundation to support effective troubleshooting and repair. It is not a comprehensive, in-depth explanation of the product architecture or technical implementation. It offers enough information on the functions and operations of the monitoring systems so that engineers who repair them are better able to understand how they work.

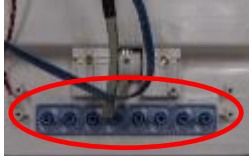
Only qualified service personnel should attempt to install the system, disassemble the monitor, remove or replace any internal assemblies, or replace the transducer cable.

3.1. *Product*

Remember to store all screws and parts in a safe place for later refitting.

Step	Picture	Description
1		Remove the four screws on the rear panel.
2		Separate the top assembly from the bottom assembly.
3	 [Bottom Ass'y]  [Top Ass'y]	Top assembly and bottom assembly.

4-1		Disjoint the printer ground screw.
4-2		Disjoint the four fixing screws which are fixing the printer assembly.
4-3		Separate the printer assembly.
5		Separate the sensor bracket.
6-1		Disjoint the seven screws which are fixing the main board.
6-2		Separate the main board and digital bracket.

7-1		Disjoint the four screws which are fixing the EMI plate.
7-2		Separate the EMI plate.
8	 	Disjoint the two screws which are fixing power switch board. Then separate power switch board and power switch.
9	 	Disjoint the four screws which are fixing key board. Then separate key board and key.
10	 	Disjoint the two screws which are fixing encoder board. Then separate encoder board.
11		
12	 	To remove hinge cover, push the locking structure using narrow driver. Then remove hinge cover.

13-1		Disjoint the two screws which are fixing LCD or LED assembly.
13-2		Separate LCD or LED assembly.
13-3		Main top cover. Main Top Cover

3.2. How to disjoint LCD Assembly

Step	Picture	Description
1		Disjoint the four screws in front and back side of assembly. Remove alarm indicator. Then separate top and bottom covers.
2	 	LCD or LED board is in the inside of top cover.

3.3. *Doppler Probe*



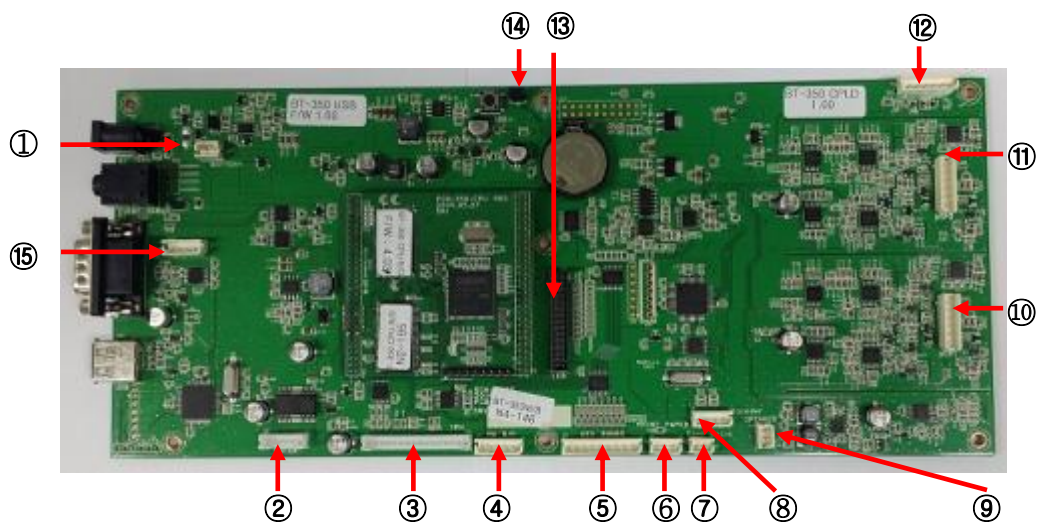
Remove the rubber protectors and disjoint the four screws. Then separate Doppler top cover from bottom cover carefully.

3.4. *UC Probe*



Remove the rubber protectors and disjoint the four screws. Then separate UC top cover from bottom cover carefully.

3.5. Connector position



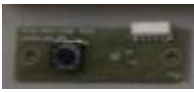
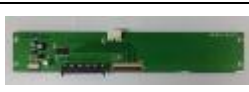
- ① Inner Battery Connector
- ② Motor Connector
- ③ TPH Connector
- ④ Power Switch Board Connector
- ⑤ Key Board Connector
- ⑥ Print Paper Connector
- ⑦ Door Print Connector
- ⑧ Encoder Board Connector
- ⑨ Speaker Connector
- ⑩ DOP1 Probe Connector
- ⑪ DOP2 Probe Connector
- ⑫ UC Probe Connector
- ⑬ LCD Board Connector
- ⑭ Jumper
- ⑮ WiFi Module Connector

Section 4

Part List

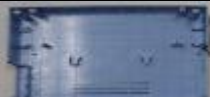
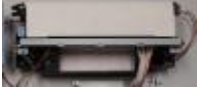
4.1. Default Parts

4.1.1. Top Ass'y Parts

No	Part No.	Part name	Picture
1	350-MP-TOP-01R01	Main Top Cover (LCD / LED shared)	
2	BD-350-PSW-R01	Power Switch Board	
3	BD-350-KEY-R00	Key Board	
4	BD-350-KNOB-R00	Knob Board	
5	SPK-8/8W	Speaker	
6	WH-350-05	Key Board Cable (9Pin)	
7	WH-350-04	Power Switch Cable (5Pin)	
8	WH-350-03	Knob Board Cable (4Pin)	
9	AY-LCD-350	LCD Assy	
10	LCD-LTP700WV-F01-0	LCD	
11	BD-350-LCD-R00	LCD Board	

13	AY-LED-350	LED Ass'y	
14	BD-350E-LED-R02	LED Board	
12	WH-350-02	Hinge Cable (with Hinge)	

4.1.2. Bottom Ass'y Parts

No	Part No.	Part name	Picture
1	350-MP-BTM-01R02	Main Bottom Cover	
2	AY-350-MAIN-R07	Main Board (with CPU Board)	
3	AY-SENSOR BRACKET-350	Sensor Bracket Ass'y	
4	350-MP-BRT-01R01	Digital Bracket	
5	350-MP-EMI-01R00	EMI Plate	
6	AY-PRINTER-350	Print Ass'y	
7	TPH-KFA-216-8GAM1-BIS	TPH	
8	MOT-PM35S-048-YKP0	Motor	

4.1.3. Accessories Parts

No	Part No.	Part name	Picture
1	350-MP-BTM-02R00	Doppler Bottom Cover Ass'y (Including Ultrasound Sensor)	
2	350-MP-BTM-03R01	UC Bottom Cover	
3	350-MP-TOP-03R00	Probe Top Cover (DOP / UC)	
4	BD-350-UC-R03	UC Board	
5	SEN-35MIT23	Strain Gauge	
6	CORD-PC-05-350-01	DOP Probe Cable	
7	CORD-PC-05-350-02	UC Probe Cable	

4.1.4. Accessory & Option Parts

Accessory	Name	Description
	Doppler Probe	Ultrasound Transducer for Measuring FHR (IPX8 : Waterproof)
	UC Probe	Pressure Sensor (Tocotonometer) for Measuring Uterine contraction (IPX8 : Waterproof)
	Event Marker	Used for a Fetal Movement event
	AST Probe (Option)	Acoustic Stimulation Test Probe
	Z-folded type Paper	Z-folder type thermal Paper
	Probe Belt	Used for Holding Doppler Probe and/or UC Probe
	Power Cord	AC Power cord
	Power Adaptor	Adaptor for transform AC Power (100-240V ~) to DC 18V(3.4A) (JMW160, Bridge Power Corp.)
	LI-ION Battery	Type / model : 18650 Technical data : 3.7V * 4ea, 2600mA
	Ultrasound Gel	Ultrasound transmission gel (Sanipia, ECOSONIC)

Section 5

External device Communication

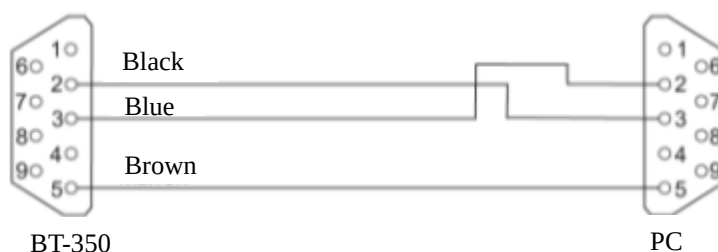
5.1. RS-232C Port

Using RS-232C cable or multi-port, you can make a wired communication between PC and BT-350.

5.1.1. 1:1 connection

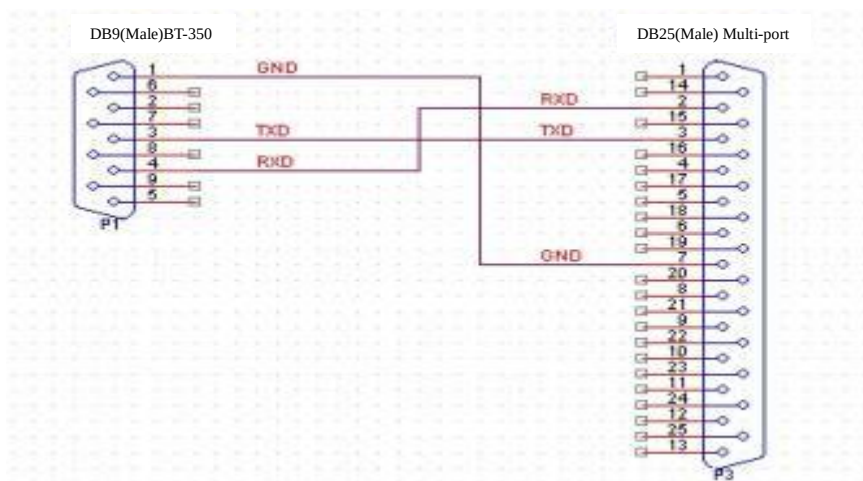
You can get RS RS-232C cross cable in the market. Both side of connector is D-SUB9 (Female). Or you can make a cable with the connection information as below.

(Please secure the enough length of cable considering setting place and distance between PC and a fetal monitor.)



5.1.2. Multi connection

In case of connecting monitors to the central monitoring PC through multi-port. The cable information is as follow.



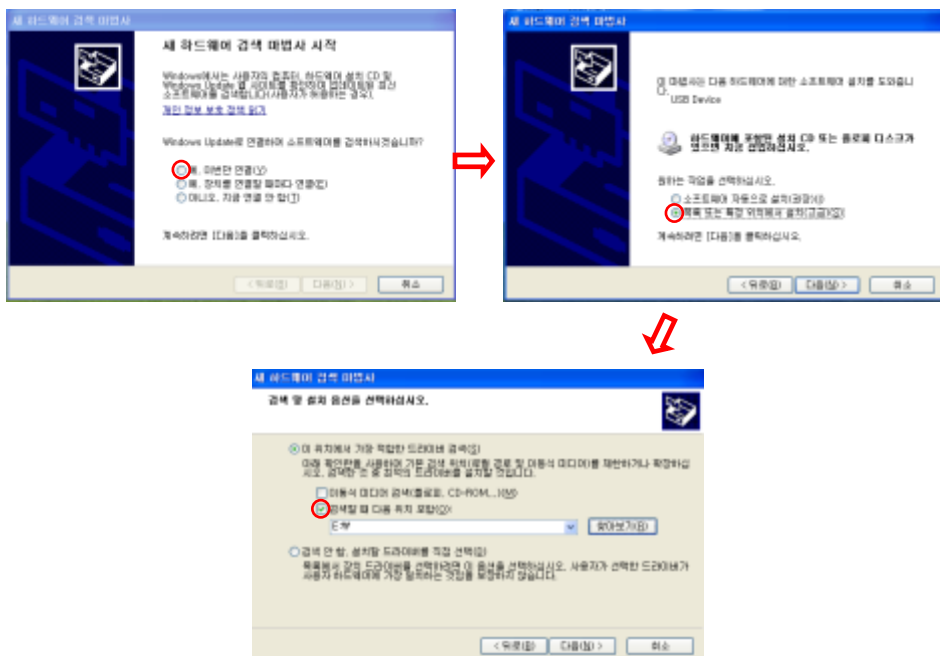
Section 6

Software Upgrade

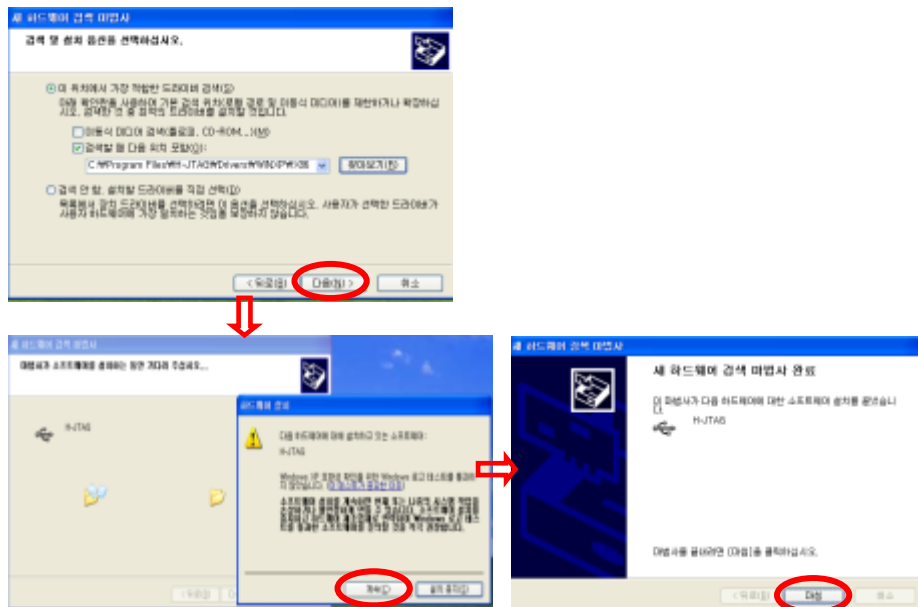
Our company Supports “ Software upgrade function “ for the algorithm or user interface of BT-350. Software upgrade function must be operated by authority field engineer. Follows sequence of Software upgrade function.

6.1. How to do S3C2410 flash fusing

- ① Open <http://www.hjtag.com/download.html>
- ② Download H-JTAG V1.1 RELEASE (Build 20100601)
- ③ Open compressed file, and install program file.
- ④ Connect H-JTAG to PC, and install USB Drivers.



- ⑤ Select red round above. Then find and select the folder which is accordance with your system. (you can find this folder generally “C:\Program Files\H-JTAG\Drivers”)



- ⑥ Connect the connector to the cable of JTAG. You can see 'J2' on the connector. It could be a reference for proper connection.



- ⑦ Remove Jumper on BT-350 Main Board.



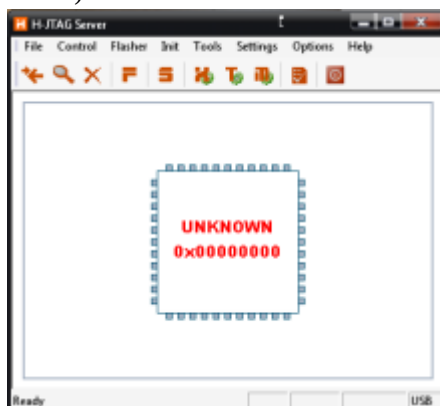
- ⑧ Connect JTAG dongle(left hand side of second image below) to J3 on BT-350 CPU module (S3C2410 CPU module) and connect USB Jack(right hand side of second image below) to PC.



- ⑨ Run H-JTAG

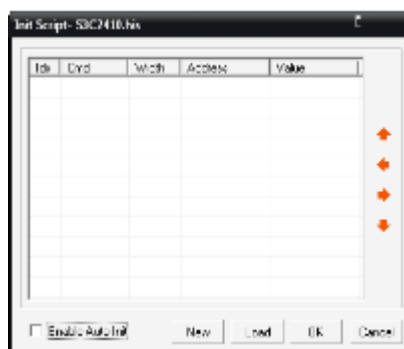
H-jtag.lnk

You can see below on the screen. Then turn on the device(Fetal monitor). TGT of JTAG lights. (If you see “ARM910T” already on the screen, pls. proceed from no. 13)



- ⑩ Click “Init” of red round above.

Init → Init script



Click “Load” and select “S3C2410.his” file, then click “OK”.

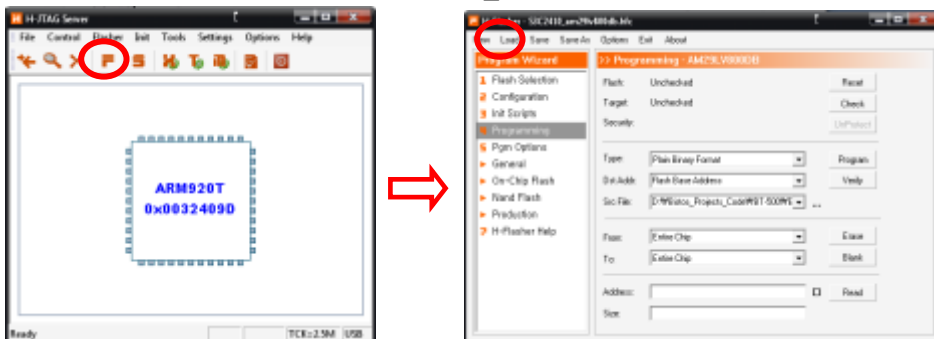
Click “Init” – “Auto init” and confirm “V” mark.



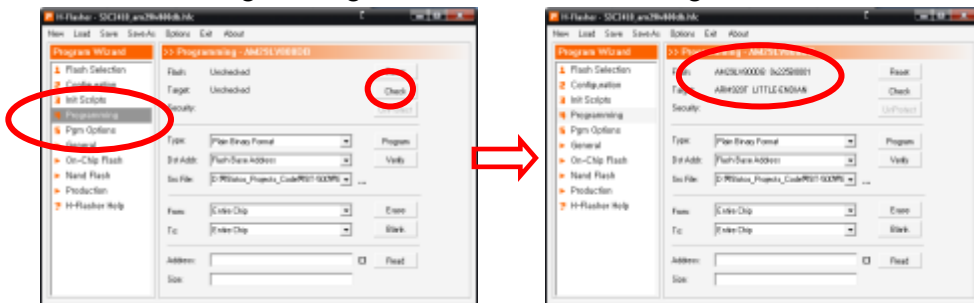
- ⑪ Click the icon in red round below image and confirm ARM920T is shown as follows.



- ⑫ Click “F” icon of red round below. Then you can see the right screen.
Click “Load” and select “S3C2410_am29lv800db.hfc” file.



- ⑬ Select “Programming”, and Click “Check” of Target.

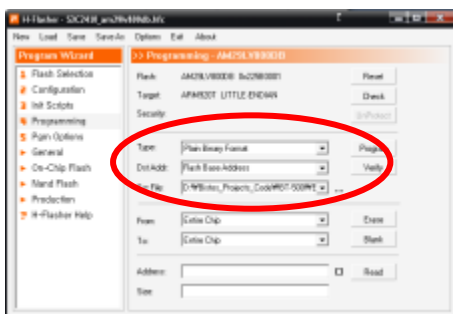


Confirm you see “AM29LV800DB 0x225b0001” and select as below.

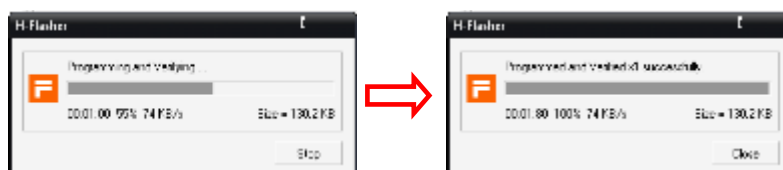
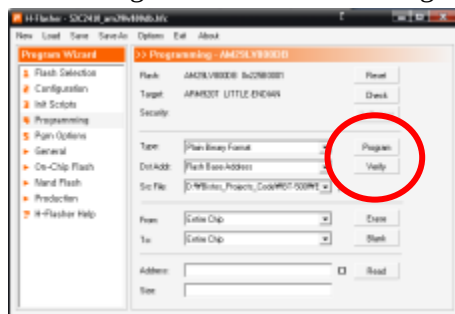
Type : Plain Binary Format

Dst Addr : Flash Base Address

Src : XXX.bin (provided file)



Click “Program” and do flash fusing.



Turn off and disconnect H-JTAG. Connect Jumper and run factory set up.

Section 7

Product Setting

7.1. Power & Accessories Connection

section	Picture	Description
Power		Connect the power adaptor to power connector.
Mark		Connect the marker jack to marker connector.
DOP1		Connect the DOP1 probe to DOP1 connector.
DOP2		Connect the DOP2 probe to DOP2 connector..
UC		Connect the UC probe to UC connector.

Section 8

Maintenance & Conservation

This chapter contains instructions for the care and cleaning of the BT-350 unit and its accessories.

The BT-350 requires proper care and preventive maintenance. This ensures consistent operation and maintains the high level of performance necessary in monitoring procedures.

8.1. Monitor

Keep the external surface clean and free of dust, dirt, and residual liquids. Clean with a damp cloth using mild soap and water or hospital approved, nonabrasive disinfectants.

WARNING: Unplug the monitor from the AC power source and detach all accessories before cleaning. Do not immerse the unit in water or allow liquids to enter the case.

CAUTION: Take extra care when cleaning the display surfaces, which are sensitive to rough handling. Rub the lens that covers them with a soft, dry cloth.

8.2. Transducers

Cleaning and Disinfecting the Tocotonometer and Ultrasound Transducer

To avoid damage to the transducers, clean and disinfect only according to the following instructions. Care **MUST** be taken to preserve both the Tocotonometer label and the Tocotonometer cable label. **DO NOT** remove, conceal or deface Tocotonometer labels.

CAUTION: Do not autoclave. Do not gas sterilize.

1. Wipe the device with a sterile wipe soaked in enzymatic detergent safe for use with metal instruments. Wipe the exterior of the device three times. Prepare the detergent according to the manufacturer's transducer recommendations.
2. Scrub the transducer with enzymatic detergent using soft bristled brush for five (5) minutes..

CAUTION: Do not immerse in liquid. When using solutions, use sterile wipes to avoid pouring fluids directly on the transducer.

3. Wipe the transducer three (3) times with sterile water to remove soap residue.
4. Wipe the transducer with a sterile wipe soaked in Cidex™. Wipe all exterior surfaces of the transducer three (3) times.

5. Wipe the transducer three (3) times with sterile water to remove Cidex residue.
6. Dry the device thoroughly with a sterile soft towel or gauze surgical sponge.
7. Wrap the dry device in a fresh sterile soft towel or transparent sterile wrap for storage until next use.

8.3. Belts

Wash soiled belts with soap and water.

CAUTION: The water temperature must not exceed 60°C (140°F).

8.4. Contacting components and characteristics

Contacting component	Material	Usage	Disinfection
DOP & UC Housing	ABS AF-302	Reusable	Must be cleaned and disinfected prior to use
Strain gauge sensor housing	RTV664	Reusable	Must be cleaned and disinfected prior to use

8.5. Description of Cidex™

1. Cidex™ is FDA-cleared for use in the United States. Therefore we suggest that the disinfection effect using Cidex™ is valid.
2. FDA-Cleared Sterilants and High Level Disinfectants with General Claims for Processing Reusable Medical and Dental Devices – March 2009

(www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/ReprocessingofSingle-UseDevices/UCM133514)

Manufacturer	Active Ingredient	Sterilant Contact Conditions	High Level Disinfectant Contact Conditions
K924434 Cidex™ Activated Dialdehyde Solution			
Johnson & Johnson Medical Products	2.4% glutaraldehyde	10 hrs at 25°C 14 days Maximum Reuse Contact conditions based on AOAC Sporicidal Activity Test only.	45 min at 25°C 14 days Maximum Reuse Contact conditions based on literature references.

8.6. Battery Disposal and Handling

CAUTION: When disposing of internal Ni-MH battery, adhere to all applicable laws regarding recycling. Avoid storing battery above 140°F. If clothing or skin

comes in contact with material from inside the battery, immediately wash with plenty of clean water.

CAUTION: The internal battery must be handled by the company's technician only. Do not attempt to open the BT-350.

The internal battery is consumables. Therefore the operation time by the battery can be decreased. If the operation time is not long enough, please contact service center and change the battery. If this system is used with not sufficient operating time by the internal battery, it is possible to be shut down the system because of the lack of the internal battery's capacity. This situation can cause not intended stop of measuring and monitoring function.

8.7. Maintenance

The BT-350 monitor and accessories require no periodic calibration or adjustment. The recommended interval for performing hipot and leakage testing is once per year.

8.8. Disposal of the BT-350

When disposing of the BT-350, adhere to all applicable laws regarding recycling. If you are not able to dispose the BT-350 or you need a help for disposing the BT-350, please contact us. In the case of there are no appropriate ways to dispose, we will pick up the BT-350 for you.

8.9. Request a service for general problems

If the main body or accessories are damaged by excessive mechanical forces, narrow cracks or separation of internal ultrasonic sensor can be happened. These can be checked through visible or auditory decision. These can cause malfunction some times. But these do not cause unacceptable risks. If the BT-350 does not work properly, please contact us and change the corresponding parts. Note that the replacement costs can be occurred.

Section 9

Troubleshooting

If the unit has trouble, check the possible cause in sequence from above.

9.1. *Warning Sign Occasion and Status Checking*

- ① After the monitor power on, Information sound rings once.
- ② If the ultrasound transducer pulls out of the connector during monitoring, the message displays right bottom of LCD, "DOP1 OPEN" and "DOP2 OPEN", it rings once every 3 seconds until the transducer is connected to the connector or pressing the  (alarm) pushbutton causes the alarm function disable.
- ③ When the  (record) pushbutton is pressed for recording in case of print door open or under recording, when the door is opened, it rings once every 3 seconds and "DOOR OPEN" in BT-350 appears on the TFT-LCD as shown below until the door is closed or pressing the  (alarm) pushbutton causes the alarm function to be disabled and return to the normal monitoring mode.
- ④ When the  (record) pushbutton is pressed for recording in case of paper empty or printer door open, or when the paper runs out under recording, it rings once every 3 seconds and "NO PAPER" in BT-350 appears on the TFT-LCD until the paper is installed or pressing the  (alarm) pushbutton causes the alarm function to be disabled and go back to the normal monitoring mode.
- ⑤ When the auto-printing period is expired, it rings once, right after stop recording.

Error Message	Cause	Solution
DOOR OPEN	Printer door open	Close the door
NO PAPER	Paper empty	Insert the paper
DOOR OPEN NO PAPER	Printer door open & Paper empty	Close the door & Insert the paper
DOP1 OPEN	Dop1 probe open under normal operation	Connect the Dop1 probe
DOP2 OPEN	Dop2 probe open under normal operation	Connect the Dop2 probe

[Figure 1.6 Errors and Trouble shooting]

● Alarm Occasion and Status Checking Up

When the measured FHR is beyond the alarm setting range for more than 20 seconds during monitoring, it rings every 3 second until the FHR is within the alarm setting range at least once or pressing the ALM(alarm) pushbutton causes the alarm function to be disabled on purpose.

9.2. Troubleshooting on the Other Situation n

9.2.1. Power and Booting Problem.

① Power adaptor.

Check Power adaptor voltage 18V.

In case low or high voltage other than 18V, or the checked voltage isn't stable, please replace Power adaptor.

② Check Power adaptor and Power Cable connection.

③ In case there is no problem on cable and power adaptor connection, please check power switch connection.

Please replace power switch board in case switch connection failure.

④ Check Main board voltage.

TP01 (18V main power)

TP02 (18V System power)

TP03 (Open)

TP04 (Battery voltage)

TP05 (VDD 5V digital voltage)

TP06 (VDD 3.3V digital voltage)

TP07 (VDD 24V Print working voltage)

TP08 (18V Power Switch input voltage)

TP09 (VDD 3.3V Clock Power voltage)

TP10 (VDD 12V analogue voltage)

TP11 (VDD 10V analogue voltage)

TP12 (AVP 6V analogue voltage)

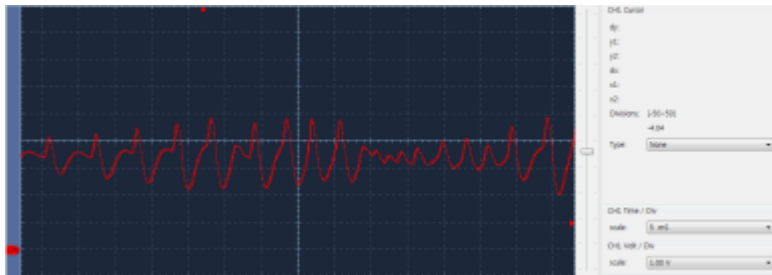
9.2.2. Doppler Probe working problem

① FHR value display problem.

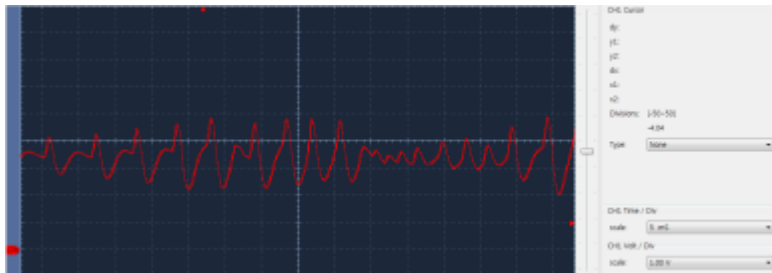
Initially please check DOP probe. In case DOP probe problem, please replace DOP probe. If there is no problem on DOP probe, please check below details.

Check Main board FHR 1 signal. You can check No. 7pin. Below oscilloscope wave is showing normal signal.

In case the signal isn't normal, please replace U31.



You can check FHR 2 signal on U34 No. 7 pin.

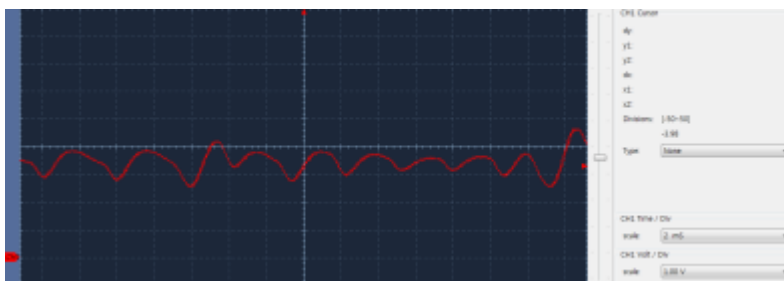


② Doppler Sound isn't coming.

Initially check speaker cable and cable connection. If there is no problem on cable and cable connection, please check below.

Check Sound signal on main board. You can check DOP1 sound signal on U30 No. 14 Pin. And you can check DOP2 sound signal on U37 No. 14 pin.

Below oscilloscope wave is showing normal signal.



Please replace IC in case it isn't showing normal signal.

③ Sound volume control problem.

Check volume signal on Main board. You can check DOP1 volume signal on

U32 No. 9, 10, 11 pin, and DOP2 on U38 No. 9, 10, 11 pin.

You can check signal changes when you press volume button. (H, L)

Volume level	Signal (No. 11, 10, 9)
0	000(Low, Low, Low)
1	001(Low, Low, High)
2	010(Low, High, Low)

9.2.3. UC Probe working problem.

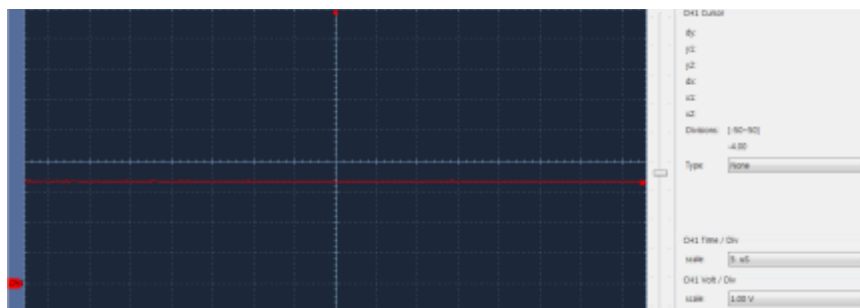
- ① UC value isn't showing on display.

Initially please check UC probe it is normal or abnormal. In case UC probe isn't normal, please replace UC probe first. If it is normal, please check followings. (It should give UC amount until 99 when you press it hardly with your finger.)

Please check below in case there is no problem on UC Probe.

Check UC signal on Main board. You can check on U25 No. 3 pin. Below oscilloscope wave is showing normal signal.

Please replace U25 in case the signal isn't normal.



② In case the problem is on UC Probe

As much as continuously uses UC Probe, there can be some changes on measured value.

In case measured value isn't correct, please follow below instruction to re-set UC probe.

Higher measured value : Reduce UC board VR2 value using +driver. (Decreasing gain value) and using VR1, set the voltage on 5V.

Lower measured value : Increase UC board VR2 value using +driver. (Increasing gain value) and using VR1, set the voltage on 5V.

After all modification, the device should give measured amount within the 조정 range of 60 ~ 80 when you put 50g scale weight on the UC probe.

9.2.4 Display problem.

① No display on LCD.

Initially check cable connection between main board and display cable.

If there is no problem on cable connection, please check followings.

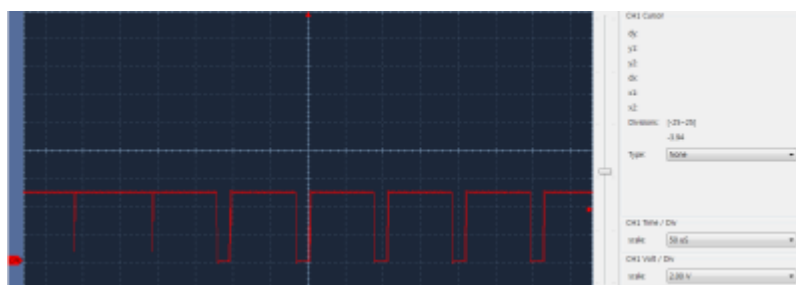
Disassemble LCD Head and check the connection between LCD and LCD board. If there is no problem on connection, please replace LCD board.

9.2.5 Alarm warning sound problem.

① In case there is problem on alarm or warning sound.

Check PWM signal on Main board. You can check on U6 No. 2 pin.

Below oscilloscope wave is showing normal signal.



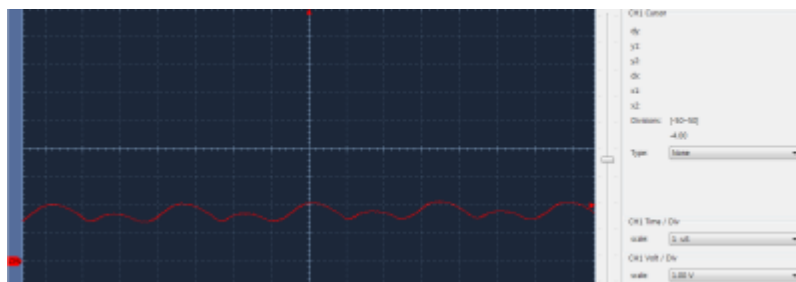
② Overall problems on sound.

Check cable connection between main board and speaker. If there is no problem on connection, please check below.

Check audio Amp out-put signal on Main board.

Check L16,L17 circuit element.

Below oscilloscope wave is showing normal signal.



9.2.6 Key working problem

① Function key working problem.

Initially please check connection between main board and Key board.

If there is no problem on connection, please check below.

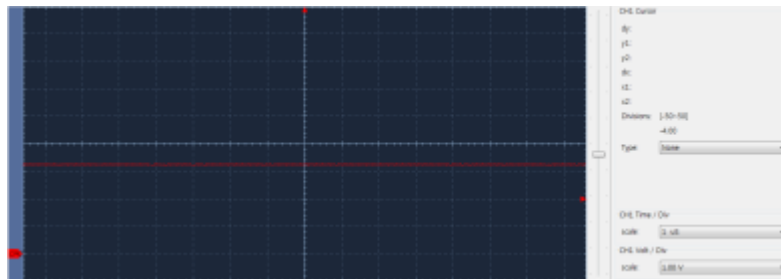
Check Key board cable connection. If there is problem on Key board cable connection, please replace Key board cable.

Check Key board switch operation statues. If there is problem, please replace Key board. If there is problem on neither Key board cable connection nor Key board, please check below.

Check Key input signal on main board. You can check on U7 No. 2~9 pin.

Below oscilloscope wave is showing normal signal.

Replace U7 in case the signal isn't normal.



9.2.7 Print working problem.

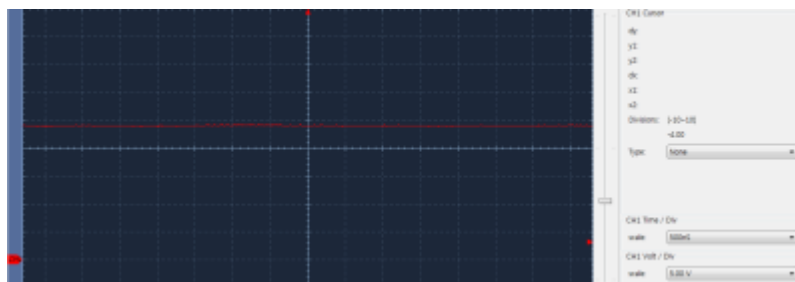
- ① Initially in case there is problem on printing, please check cable connections between printer module and main board. (TPH, Motor, Door Sensor, Paper Sensor cable connections)
- ② Print problem.

In case there is no print on particular area on printed paper, please replace인쇄 TPH(Thermal Printer Head) on print module. And if the printing quality isn't good enough on particular area on printed paper, please replace printer module, or printer body and TPH top cover.

In case the printing quality isn't good enough on all over area, or in case there is nothing printed on recording paper, or if there is printing error(misprint), please check below details.

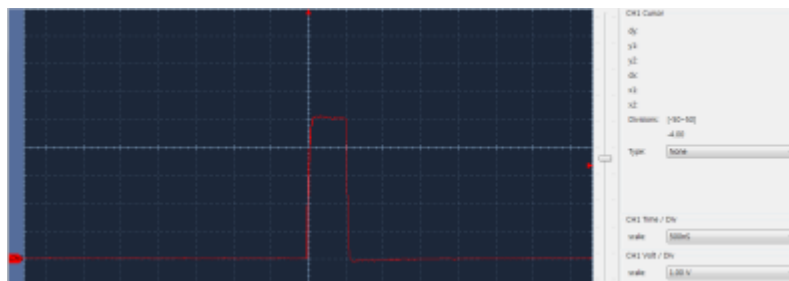
Please check Main board Print control signal.

Check voltage on L31 24V is coming.

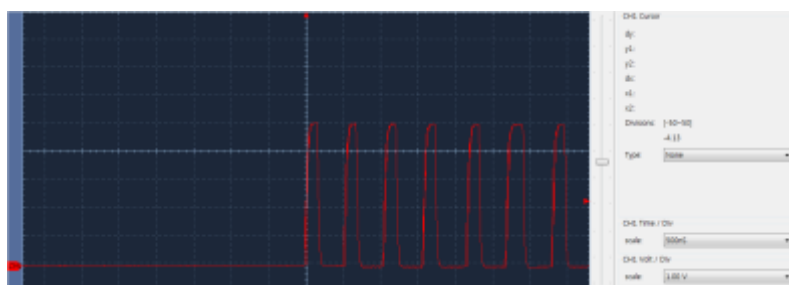


And check on U12 No. 8, 10, 12 pin, whether you can check signal as

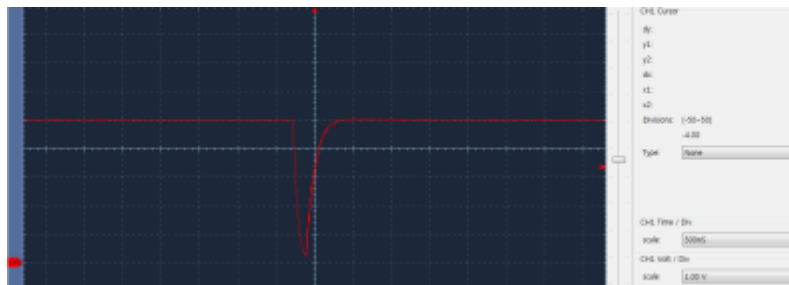
below picture.



<No.8pin>



< No.10pin >



< No.12pin >

If there is problem on signal, please replace U12.

③ Printing motor problem.

Initially please check motor gear No. 1, 2, 3. (It should rotates smoothly with finger) If there is problem on operation, please replace printing motor and motor gear. (Thick motor gear rotation, and thick rotation on particular point) If there is no problem on motor and motor gear, please check below.

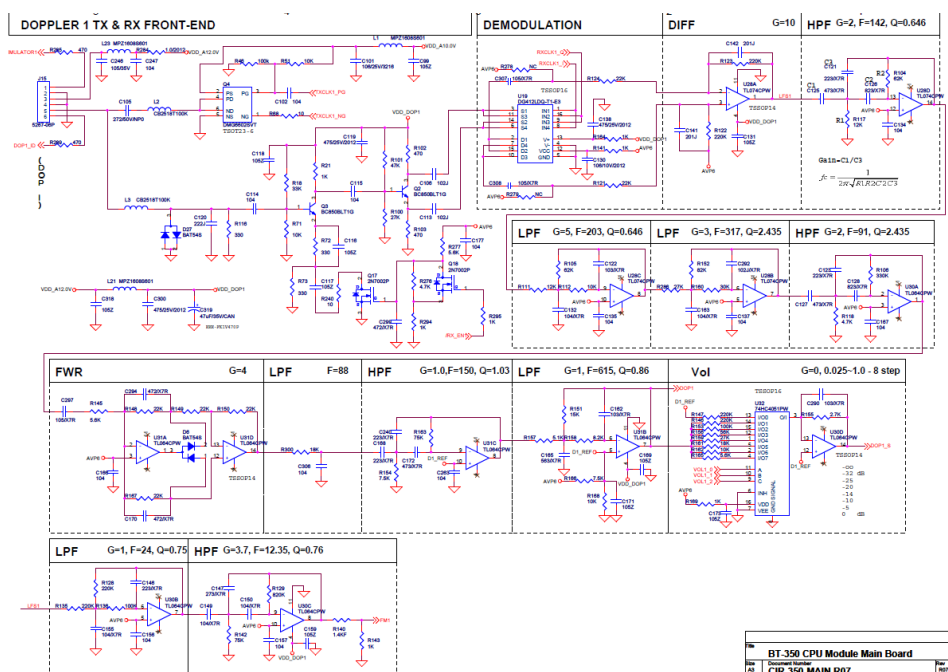
Measuring fetal heart rate is using 1MHz ultrasound wave which transmits the burst pulse to the 7 ultrasound sensors about 3.47KHz periodically. This wave is reflected to the part of different medium. Particularly, the reflection wave is shifted from moving targets like heart, blood, etc. This phenomenon is a Doppler effect. This reflected signal is converted to electrical signal and it can get some useful information by many other processes.

Also, BT-350 measures the uterine contraction of a pregnant woman by pressure sensors and displays waveform by means of numerical values. This waveform informs the period and magnitude of the uterine contraction.

9.3.1 Main board

● Circuit Diagram

<Analog Part>

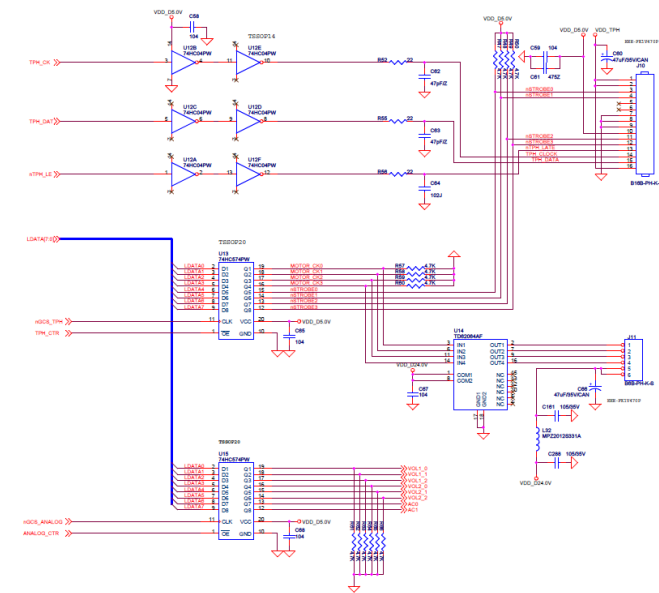


[illegible]

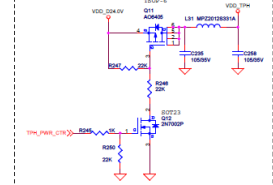
Figure 1 shows the pin connections for the KeyBoard module. The left diagram illustrates the 8-bit keyboard module, where pins 1 through 8 are connected to KEY0 through KEY7. The right diagram illustrates the 16-bit keyboard module, where pins 1 through 8 are connected to KEY0 through KEY7, and pins 9 through 16 are connected to KEY8 through KEY15. Both diagrams show the module connected to a 5V supply and ground.

<Print part>

Thermal Printer

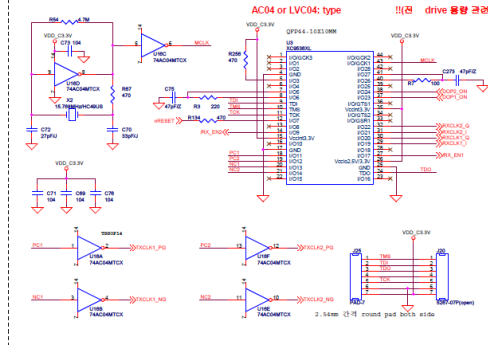


TPH Power Control

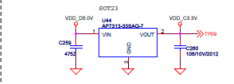


<Pulse part>

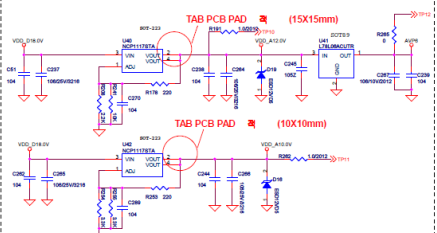
DOPPLER CLOCK GENERATION BLOCK



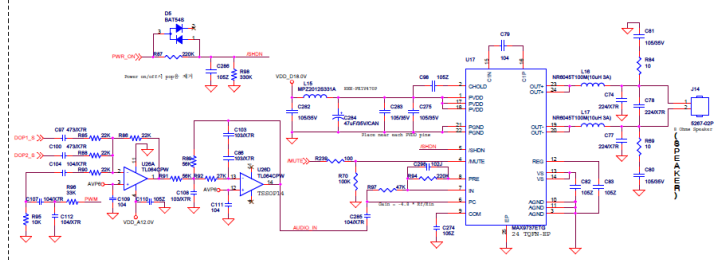
CLOCK POWER



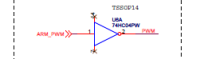
ANALOG POWER



AUDIO AMP BLOCK



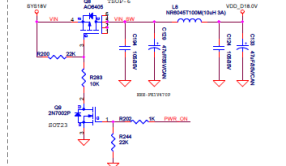
PWM



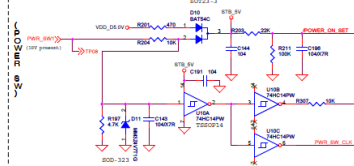
BT-350 CPU Module Main Board
CIR-350.MAIN R07

<Power>

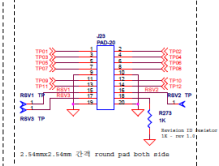
SOFT START POWER SWITCH



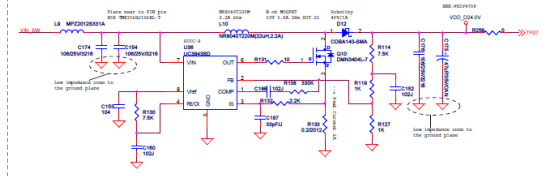
SOFT START CONTROL



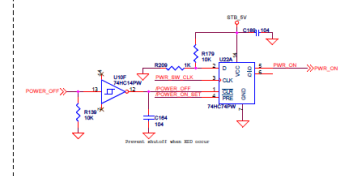
POWER TEST PORT



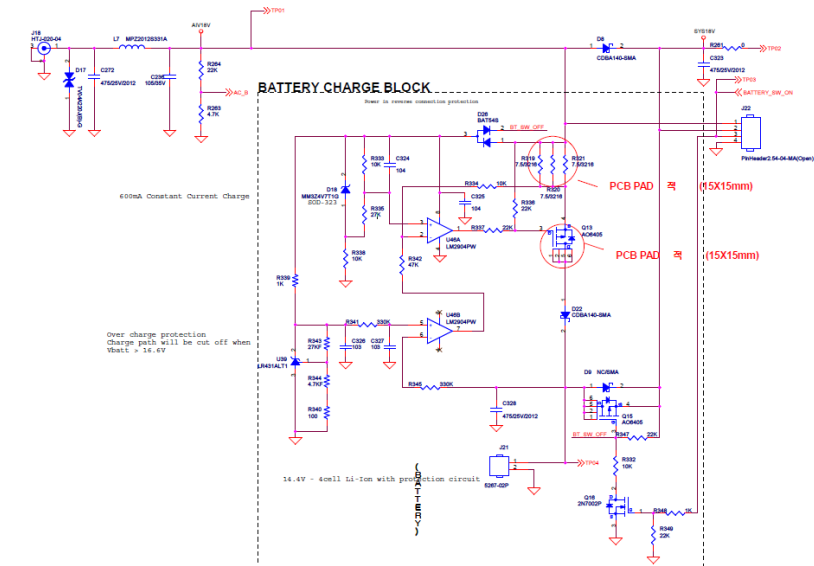
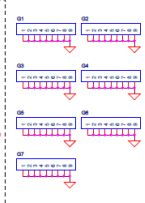
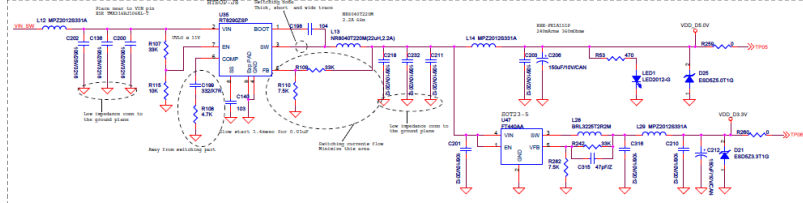
24V.TPH.& MOTOR POWER BLOCK



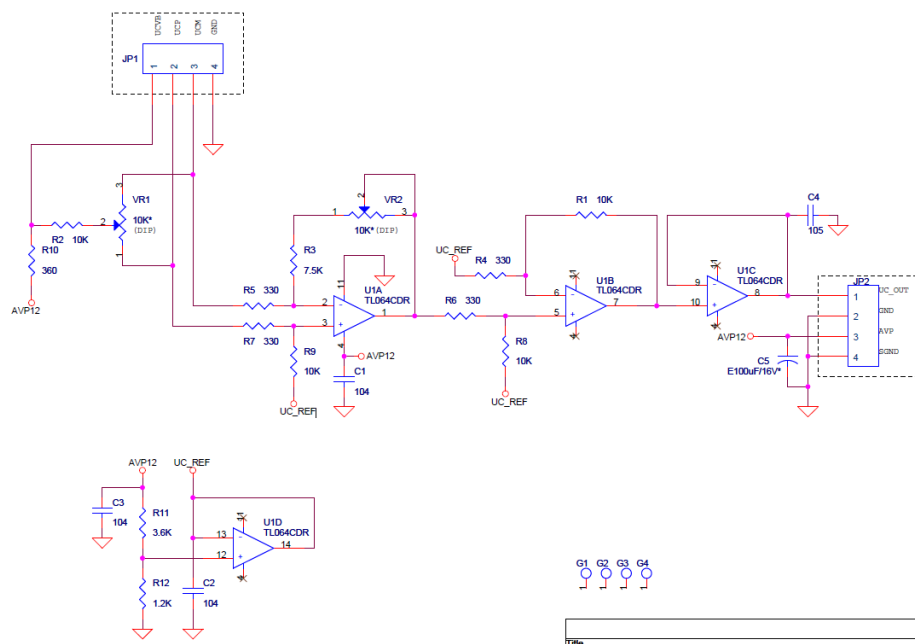
SOFT START CONTROL



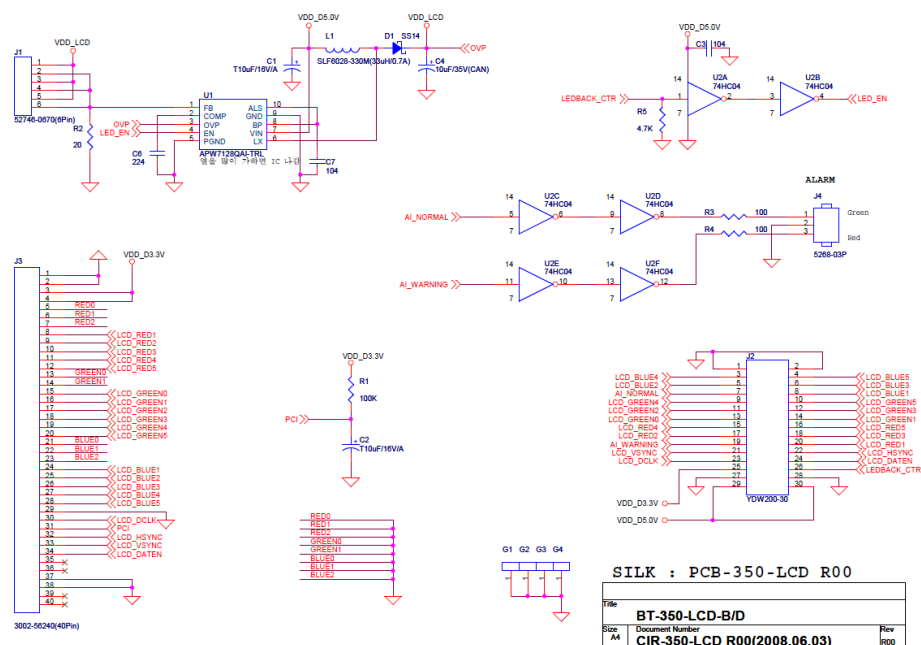
DIGITAL POWER BLOCK



<UC part>



<LCD Board part>



SILK : PCB-350-LCD R00

File	BT-350-LCD-B/D	Rev	R00
Doc Number	CIR-350-LCD R00(2008.06.03)		

① Doppler signal generation circuit

The burst signal drives the ultrasound sensor using by Q4.

The L-C impedance matching circuit is used to match the received ultrasound signal and the signal is amplified about 40 dB at Q7.

U19 demodulates by 1MHz to get the Doppler signal and the most adequate signals are abstracted through the 4 stages filter.

BT-350 uses the 1MHz ultrasound, so the Doppler frequency is low frequency about 100 ~ 200Hz. The OP Amps (U31A, U31D) perform the frequency doubling by full-wave rectifier to hear clearly when user hears the fetal heart signal. After using the amplifier and LPF, this signal is adjusted the volume (U32, 8 steps) and transferred to audio amplifier (U17).

The rectifier circuit (U31A, U31D) is used to calculate the heart rate, this signal is amplified and used LPF for envelope detection. And adjust the DC level to fit in the input level of U25, A/D converter (0 ~ 3.3V).

② Sound output circuit

The signal of Doppler (DOP1_S, DOP2_S), alarm and information are summed in U26A. The signal for alarm(bee bee bee) and information(ding~ dong~) is PWM signal generated from U43.

This signal is converted to be able to hear using RC low pass filter. This signal and Doppler signal are integrated and output to speaker by audio amplifier (U17).

③ A/D converter & ARM core circuit

MCP3204(U25) A/D converter is possible to convert to 4 channels and converts each signal DOP_1HR, DOP2_HR, UC to 12 bits digital signal by 200Hz periodically. The range of analog input signal of this A/D converter must be in 0 ~ +3.3V. The converted digital signal is transmitted to ARM core(U1) by serial and calculated an useful value using many signal

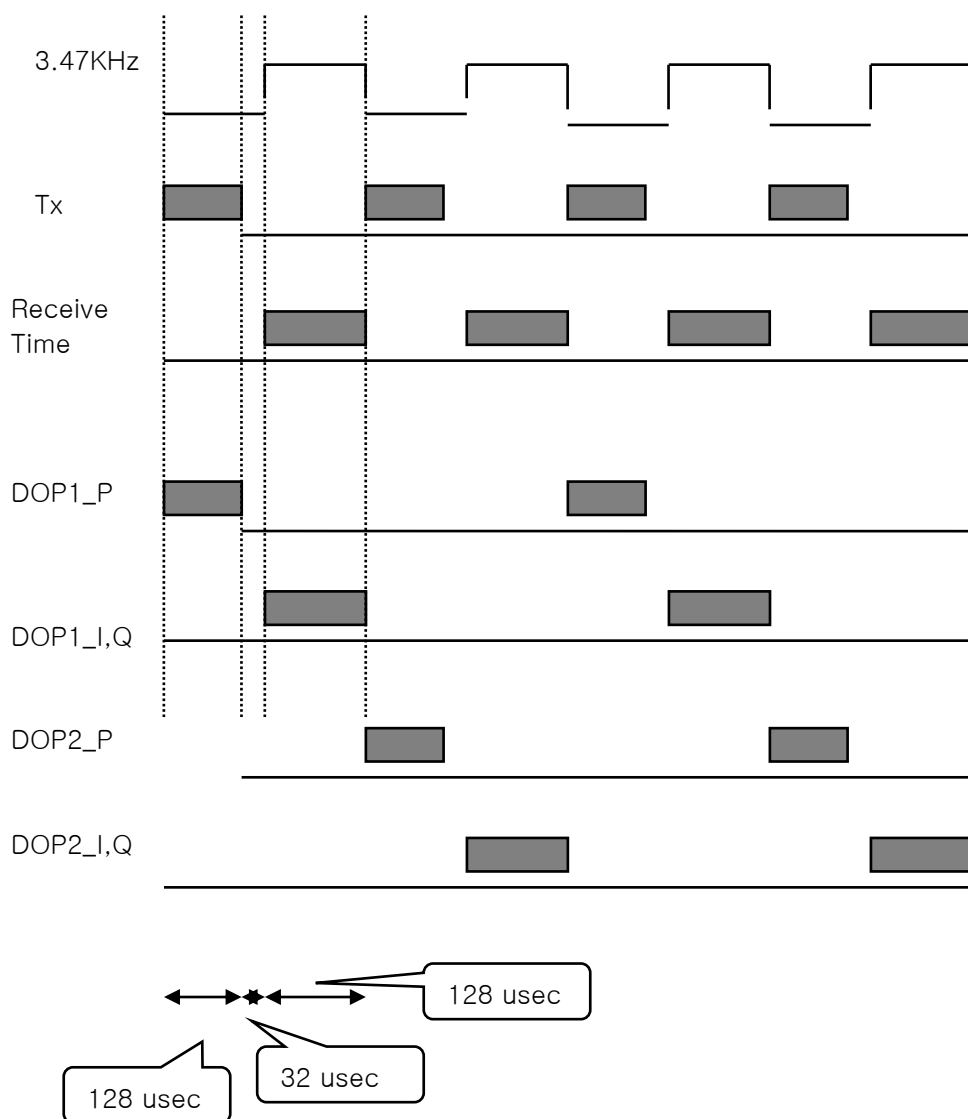
processing techniques. DSP is S3C2410 by samsung company and the fetal heart rate is calculated at 0.25 second using auto-correlation algorithm. The output is transmitted to CPU to display and record.

④ TX burst pulse generation circuit

U3 generate the burst signal for driving the ultrasound sensor and control signal for demodulation.

The timing diagram is below.

< Timing Diagram >



⑤ CPU circuit

CPU(U1) is S3C2410, when it has malfunction, CPU resets itself using CAT1163(U2) watch-dog IC. FHR and UA are displayed on the signal quality is displayed on the LCD.

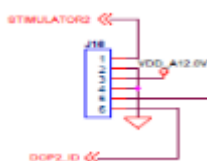
Thermal print is operated +24V. CPU controls the stepping motor and TPH to record the calculated data on the thermal paper. Door open or paper out is recognized by sensor. If that situation happens during recording, then CPU stops recording and displays “See error message” on the display panel.

Time and date are calculated automatically at the DS1337(U5). CPU reads the value and records on the recording paper.

Each key is recognized by CPU which performs each operation.

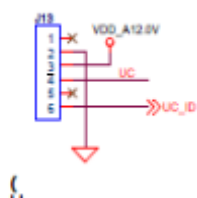
● Connection

A. Doppler Probe Connector



Pin	Name	I/O	Description
1	Stimulator	I	Acoustic stimulator enable signal
2	GND	-	Ground
3	AVP12	O	+12V power
4	GND	-	Ground
5	Signal	I/O	Transmitting and receiving ultrasound signal
6	DOP_ID	I	Identity signal

B. UC Probe Connector



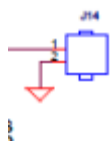
Pin	Name	I/O	Description
1	NC	-	-
2	GND	-	Ground
3	AVP12	O	+12V power
4	Signal	I	UC out signal
5	NC	-	-
6	UC_ID	I	Identity signal

C. Mark Jack Connection



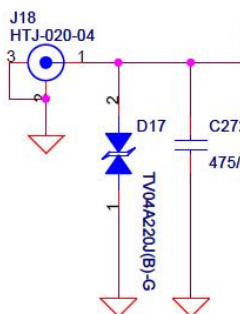
Pin	Name	I/O	Description
1	Signal	I	Mark enable signal
2	GND	-	Ground

D. Speaker Connector



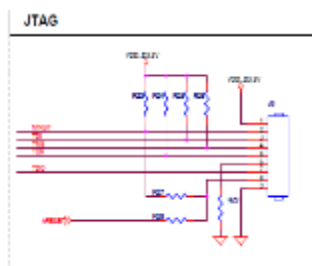
Pin	Name	I/O	Description
1	Signal	O	Sound output
2	GND	-	Ground

E. Input Power Connector



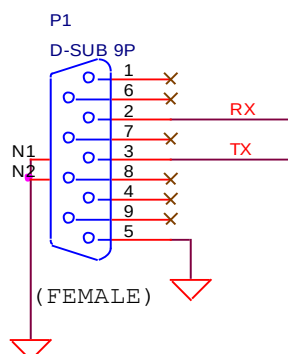
Pin	Name	I/O	Description
1	IVP	I	+18V input power
2	IVP16	I	+18V input power through power switch

F. CPU Program Download Connector



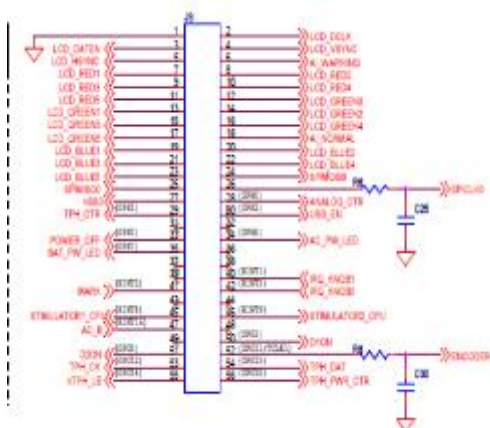
This connector is used only to download the CPU program.

G. Central Monitor Interface Connector (RS-232C)



Pin	Name	I/O	Description
2	RX	I	Receiving signal for central monitor
3	TX	O	Transmitting signal for central monitor
5	GND	-	Ground

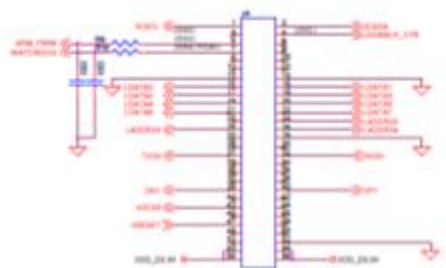
H. CPU Board Connector



Pin	Name	I/O	Description
1	GND	O	LCD Display signal
2	LCD_DCLK	O	LCD Display signal
3	LCD_Daten	O	LCD Display signal
4	LCD_VSYNC	O	LCD Display signal

5	LCD_HSYNC	O	LCD Display signal
6	AI-WARNING	O	LCD Display signal
7	LCD_RED1	O	LCD Display signal
8	LCD_RED2	O	LCD Display signal
9	LCD_RED3	O	LCD Display signal
10	LCD_RED4	O	LCD Display signal
11	LCD_RED5	O	LCD Display signal
12	LCD_GREEN0	O	LCD Display signal
13	LCD_GREEN1	O	LCD Display signal
14	LCD_GREEN2	O	LCD Display signal
15	LCD_GREEN3	O	LCD Display signal
16	LCD_GREEN4	O	LCD Display signal
17	LCD_GREEN5	O	LCD Display signal
18	AI-NORMAL	O	LCD Display signal
19	LCD_BLUE1	O	LCD Display signal
20	LCD_BLUE2	O	LCD Display signal
21	LCD_BLUE3	O	LCD Display signal
22	LCD_BLUE4	O	LCD Display signal
23	LCD_BLUE5	O	LCD Display signal
24	SPIMOSI0	O	A/D Convertor control signal
25	SPIMOSO0	I	A/D Convertor control signal
26	SPICLK0	O	A/D Convertor control signal
27	nSS0	O	A/D Convertor control signal
28	ANALOG_CTR	O	Analog control signal
29	TPH_CTR	O	Printer control signal
30	USB_EN	O	USB control signal
31	NC		
32	NC		

33	POWER_OFF	O	Power Off control signal
34	AC_PW_LED	O	AC Power led control signal
35	BAT_PW_LED	O	Batteries Power led control signal
36	NC		
37	NC		
38	NC		
39	NC		
40	IRQ_KNOB1	I	Knob Key control signal
41	MARK	I	Mark Input signal
42	IRQ_KNOB0	I	Knob Key control signal
43	NC		
44	NC		
45	STIMULATOR1_CPU1	O	Stimulator1 Input signal
46	STIMULATOR1_CPU2	O	Stimulator Input signal
47	AC_B	I	Power control signal
48	NC		
49	NC		
50	D1ON	O	DOP1 control signal
51	D2ON	O	DOP2 control signal
52	ENCODER	O	Encoder control signal
53	TPH_CK	O	Printer control signal
54	TPH_DAT	O	Printer control signal
55	nTPH_LE	O	Printer control signal
56	TPH_PWR_CTR	O	Printer control signal

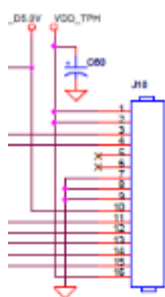


Pin	Name	I/O	Description
1	IISCL	O	Reset control signal
2	IICSDA	O	Reset control signal
3	NC		
4	LEDBACK_CTR	O	LCD Back light signal
5	ARM_PWM	O	CPU control signal
6	NC		
7	WATCHDOG	O	Reset control signal
8	NC		
9	NC		
10	NC		
11	NC		
12	NC		
13	GND	O	Ground
14	GND	O	Ground
15	LDATA0	O	CPU Data signal
16	LDATA1	O	CPU Data signal
17	LDATA2	O	CPU Data signal
18	LDATA3	O	CPU Data signal
19	LDATA4	O	CPU Data signal
20	LDATA5	O	CPU Data signal
21	LDATA6	O	CPU Data signal

22	LDATA7	O	CPU Data signal
23	NC		
24	LADDR22	O	CPU Data signal
25	LADDR23	O	CPU Data signal
26	LADDR24	O	CPU Data signal
27	NC		
28	GND		Ground
29	NC		
30	NC		
31	TXD0	O	CPU Control signal
32	RXD0	I	CPU Control signal
33	NC		
34	NC		
35	NC		
36	NC		
37	NC		
38	NC		
39	DN1	O	CPU Control signal
40	DP1	O	CPU Control signal
41	NC		
42	NC		
43	nGCS3		CPU Control signal
44	NC		
45	NC		
46	NC		
47	nRESET	I	Reset control signal
48	NC		
49	NC		

50	NC		
51	NC		
52	GND		
53	VDD_D3.3V	O	3.3V Power signal
54	VDD_D3.3V	O	3.3V Power signal
55	VDD_D3.3V	O	3.3V Power signal
56	VDD_D3.3V	O	3.3V Power signal

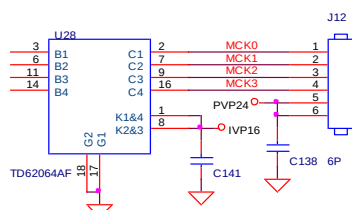
I. TPH Connector



Pin	Name	I/O	Description
1	PVP24	O	+24V power
2	PVP24	O	+24V power
3	STROBE0	-	TPH strobe 0 signal
4	STROBE1	-	TPH strobe 1 signal
5	NC		
6	NC		
7	GND		Ground
8	GND		Ground
9	GND		Ground
10	PVP5	O	+5 power
11	STROBE2	O	TPH strobe 2signal

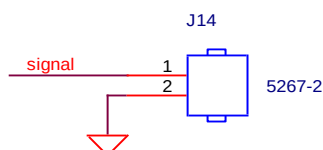
12	STROBE3	O	TPH strobe 3signal
13	nTPH_LE	O	TPH LE signal (active low)
14	TPH_CLOCK	O	TPH clock signal
15	TPH_DATA	O	TPH data signal
16	PVP24		

J. Motor Connector



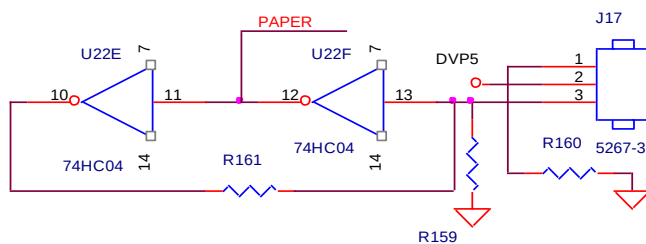
Pin	Name	I/O	Description
1	MCK0	O	Motor clock 0 signal
2	MCK1	O	Motor clock 1 signal
3	MCK2	O	Motor clock 2 signal
4	MCK3	O	Motor clock 3 signal
5	PVP24	O	+24V power
6	PVP24	O	+24V power

K. Printer Door Check Connector



Pin	Name	I/O	Description
1	SIGNAL	I	Door open check signal
2	GND	-	Ground

L. Printer Paper Check Connector



Pin	Name	I/O	Description
1	LL	O	Level signal
2	DVP5	O	+5V power
3	PAP	I	Paper check signal

Service Telephone and Fax. Numbers

Telephone: +82 31 750 0340

Fax: +82 31 750 0344

Manufacturer: Bistos Co., Ltd.

7th Fl., A Bldg., Woolim Lions Valley 5-cha,
302, Galmachi-ro, Seongnam-si, Gyeonggi-do, Korea

www.bistos.co.kr

bistos@bistos.co.kr

Model Name: BT-350

EC Representative: Medical Econet GmbH

Im Erlengrund 20 D-46149 Oberhausen Germany

Telephone: +49 (0)208 377 890 22

Fax: +49 (0)208 377 890 55

